Report on

SUPPLY CHAIN AND LOGISTICS ANALYTICS

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1. Introduction

1.1 Background of Study

Supply Chain Management involves the planning and coordination of activities related to products, inventory, manufacturing, transportation, and sales. Modern organizations generate large amounts of supply chain data, making data analysis important for understanding business performance and improving operational efficiency.

The SupplyChain360: Integrated Supply Chain Analytics Dataset was used in this project to analyze supply chain activities through Microsoft Power BI. The dataset contains 15,000 supply chain transaction records along with supporting Product, Location, and Date tables.

Power BI was used to transform the raw data into an interactive dashboard containing KPIs, charts, filters, and business insights. The dashboard provides information about revenue, profit, orders, profit margin, inventory availability, defect rate, supplier lead time, product sales, location performance, and supply chain costs.

The project demonstrates how Business Intelligence and data visualization can help organizations understand supply chain performance and support data-driven decision-making.

1.2 Problem Statement

Supply chain data contains information from different business activities, making it difficult to understand overall performance using raw datasets or traditional reports.

The main problem is to organize and analyze supply chain transaction data in a structured way so that important business information such as revenue, profit, product demand, costs, inventory availability, and operational performance can be easily understood.

Therefore, the project aims to develop an interactive Power BI dashboard that converts raw supply chain data into meaningful visual insights and Key Performance Indicators.

1.3 Objectives of the Project

The main objectives of the project are:

1. To analyze the performance of the supply chain using Power BI.
2. To organize the dataset into a structured multi-table data model.
3. To create relationships between the Fact and Dimension tables.
4. To calculate important KPIs using DAX.
5. To analyze revenue, profit, orders, costs, inventory, and product sales.
6. To identify important trends and patterns in the supply chain data.
7. To create an interactive dashboard using charts, KPI cards, and filters.

8. To support data-driven decision-making through visual analysis.

1.4 Scope and Limitations

Scope

The project focuses on analyzing supply chain transaction data using Power BI. The analysis covers:

- Revenue and profit analysis
- Product sales analysis
- Location-wise revenue
- Supply chain cost analysis
- Inventory availability
- Defect rate
- Supplier lead time
- Revenue trends over time
- Interactive filtering using Product Type, Location, and Date

The project uses a Star Schema consisting of:

- SupplyChain
- Product
- Location
- Date

Limitations

1. The dataset contains 15,000 transaction records and is designed for analytical purposes.
2. The calculated profit is a profit proxy based on the costs available in the dataset.
3. Other business expenses such as salaries, taxes, marketing, and administration are not included.
4. The dashboard provides high-level analysis and does not cover every detailed supply chain operation.
5. The results are dependent on the accuracy and structure of the available dataset.

2. Literature Review / Theoretical Background

2.1 Business Intelligence Overview

Business Intelligence (BI) refers to the use of technologies, processes, and analytical methods to collect, transform, analyze, and present business data.

Business Intelligence helps organizations convert raw data into meaningful information that can be used for planning and decision-making.

In supply chain management, Business Intelligence can be used to analyze:

- Sales and revenue
- Inventory
- Product demand
- Manufacturing costs
- Transportation costs
- Supplier performance
- Quality performance
- Customer demand
- Operational trends

Power BI is one of the widely used Business Intelligence tools that provides data transformation, data modelling, DAX calculations, interactive visualizations, dashboards, and reporting capabilities.

In this project, Power BI was used to analyze the SupplyChain360 dataset and create an executive-level supply chain dashboard.

2.2 Review of Related Tools

Several tools are available for Business Intelligence and data analysis.

Microsoft Power BI

Power BI is a Business Intelligence and data visualization platform developed by Microsoft. It supports data transformation, data modelling, DAX calculations, interactive reports, dashboards, and filters.

Power BI was selected for this project because it provides strong data modelling and visualization capabilities.

3. Methodology and Data Description

3.1 Data Sources

The main data source used for this project is the SupplyChain360: Integrated Supply Chain Analytics Dataset.

The dataset was expanded and transformed into a relational structure containing 15,000 supply chain transaction records.

The final model contains:

Table	Description
SupplyChain	Main transaction and performance data
Product	Product and product type information
Location	Location and regional information
Date	Date and time-related information

The Fact table contains transaction-level information such as:

- TransactionID
- DateID
- ProductID
- LocationID
- ProductsSold
- RevenueGenerated
- StockLevels
- OrderQuantity
- ShippingTime
- ShippingCost
- ManufacturingCost
- ManufacturingLeadTime
- DefectRate
- TransportationCost

- Availability
- SupplierLeadTime

3.2 Data Cleaning and Transformation

Before creating the Power BI dashboard, the dataset was prepared and structured for analysis.

The major data preparation steps included:

1. **Checking Dataset Structure**
   The available tables, columns, data types, and transaction records were reviewed.
2. **Checking Missing Values**
   The dataset was checked for blank or missing values.
3. **Checking Data Types**
   Numerical, text, and date fields were reviewed and assigned appropriate data types.
4. **Creating Dimension Tables**
   Product, Location, and Date information were organized into separate dimension tables.
5. **Creating the Fact Table**
   Transaction-level supply chain information was organized into the SupplyChain table.
6. **Creating Unique Keys**
   IDs such as ProductID, LocationID, and DateID were used to connect the tables.
7. **Data Transformation**
   Required fields were formatted and prepared for Power BI analysis.
8. **Data Validation**
   The transformed data was checked to ensure that the values and relationships were suitable for analysis.

4. Multi-Table Modeling (Relationships)

4.1 Data Schema

The project uses a Star Schema for organizing the supply chain data.

The Star Schema consists of one central fact table connected to multiple dimension tables.

Fact Table: SupplyChain

This table contains the main transaction-level data and measurable values such as revenue, products sold, costs, inventory, defect rate, and lead time.

Dimension Tables: Product, Location, Date

4.2 Relationships and Cardinality

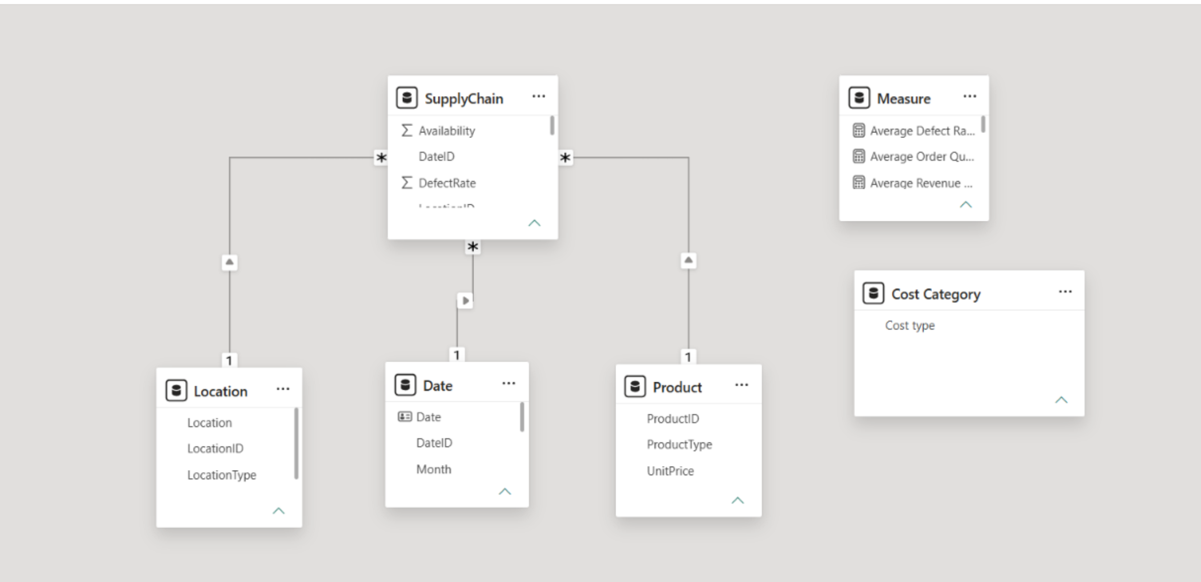
Relationships were created between the dimension tables and the central fact table.

The relationships use a **one-to-many (1:*) cardinality**.

Dimension Table	Fact Table	Primary Key	Fact Table Key	Cardinality
Product	SupplyChain	ProductID	ProductID	1 : *
Location	SupplyChain	LocationID	LocationID	1 : *
Date	SupplyChain	DateID	DateID	1 : *

Single-direction filtering is used from the dimension tables toward the fact table.

This multi-table model allows Power BI to efficiently analyze the same transaction data from different perspectives, such as product type, location, and time.



5. DAX – Calculated Columns and Measures

DAX, or Data Analysis Expressions, was used to create calculated values and KPIs in Power BI.

5.1 Total Revenue

```
Total Revenue = SUM(SupplyChain[RevenueGenerated])
```

5.2 Total Supply Chain Cost

```
Total Supply Chain Cost = [Total Shipping Cost] + [Total Manufacturing Cost] + [Total Transportation Cost]
```

5.3 Total Profit

```
Profit = [Total Revenue] - [Total Supply Chain Cost]
```

5.4 Profit Margin

```
Profit Margin = DIVIDE([Profit], [Total Supply Chain Cost])
```

5.5 Total Orders

```
Total Orders = COUNTROWS(SupplyChain)
```

5.6 Total Products Sold

```
Total Products Sold = SUM(SupplyChain[ProductsSold])
```

5.7 Average Defect Rate

```
Average Defect Rate = AVERAGE(SupplyChain[DefectRate])
```

5.8 Total Shipping cost

```
Total Shipping Cost = SUM(SupplyChain[ShippingCost])
```

5.9 Total Manufacturing Cost

```
Total Manufacturing Cost = SUM(SupplyChain[ManufacturingCost])
```

5.10 Inventory Availability %

```
Inventory Availability % =
DIVIDE(
    CALCULATE(
        COUNTROWS(SupplyChain),
        SupplyChain[Availability] = 1
    ),
    [Total Orders],
    0
)
```

5.11 Total Transportation Cost

```
Total Transportation Cost = SUM(SupplyChain[TransportationCost])
```

6. Data Analysis and Dashboard Design

6.1 Dashboard Objective

The main dashboard was designed as an Executive Overview that provides a quick summary of supply chain performance.

The dashboard focuses on:

- Revenue
- Profit
- Orders
- Profit Margin
- Revenue trends
- Location performance
- Supply chain costs
- Product demand
- Defect rate
- Inventory availability
- Supplier lead time

6.2 KPI Cards

Total Revenue

₹341M. The total revenue generated from the 15,000 transactions is approximately ₹341 million.

Total Profit

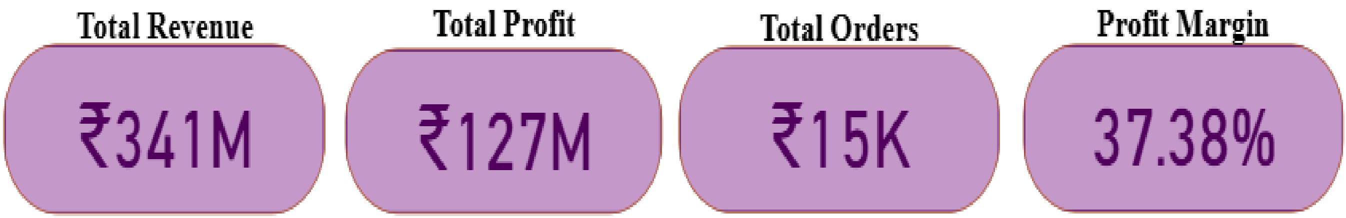
₹127M. The calculated profit proxy is approximately ₹127 million.

Total Orders

15K. The dataset contains 15,000 transaction records.

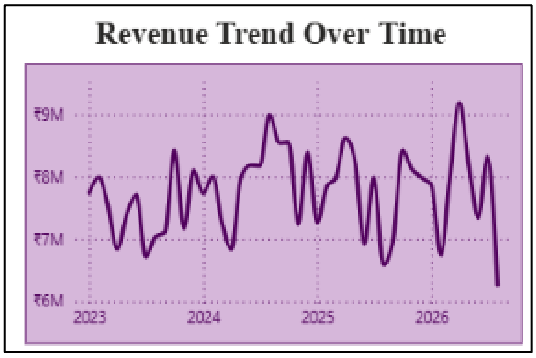
Profit Margin

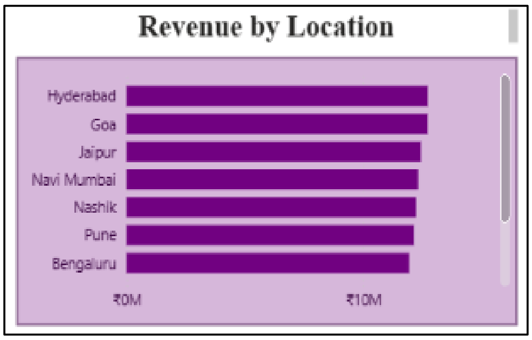
37.38%The calculated profit margin is approximately 37.38%.



6.3 Revenue Trend Over Time

A line chart is used to display revenue trends from 2023 to 2026. The chart helps identify changes and fluctuations in revenue over time.





6.4 Revenue by Location

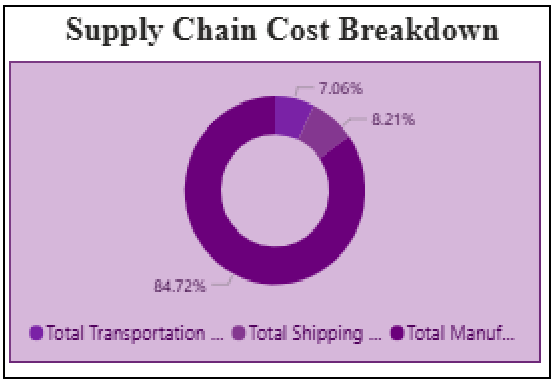
A horizontal bar chart displays revenue generated across different locations. This visualization helps compare location performance.

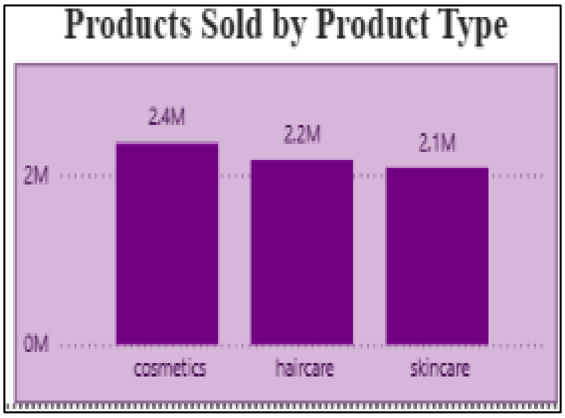
6.5 Supply Chain Cost Breakdown

A donut chart is used to display the contribution of different supply chain costs.

- Manufacturing Cost: 84.72%
- Shipping Cost: 8.21%
- Transportation Cost: 7.06%

Manufacturing represents the largest portion of the recorded supply chain cost.





6.6 Products Sold by Product Type

A bar chart compares the number of products sold by product category.

- Cosmetics: 2.38M units
- Haircare: 2.19M units
- Skincare: 2.09M units

Cosmetics has the highest product sales among the three displayed product types.

6.7 Operational KPIs

Average Defect Rate: 9.98%.

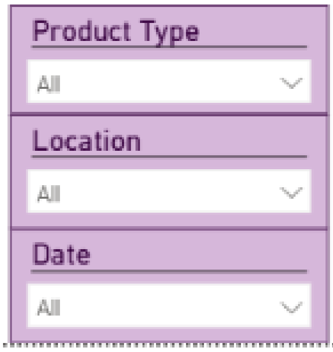
Inventory Availability: 92.41%.

Average Supplier Lead Time: 17.45 time units as represented in the dataset.

6.8 Interactive Filters

Three slicers are provided on the dashboard: Product Type, Location, and Date.

These filters allow users to dynamically change the dashboard analysis.



7. Findings and Limitations

Aspect	Issue / Finding	Recommendation
Revenue	Revenue is **₹341M**, showing strong overall sales performance.	Monitor revenue trends to maintain growth.
Profit	Profit is approximately **₹127M** with a **37.38%margin**.	Control costs to maintain profitability.
Manufacturing Cost	Manufacturing contributes **84.72%** of total supply chain cost.	Focus on reducing manufacturing expenses.
Inventory	Inventory availability is **92.41%**, indicating good product availability.	Maintain optimal inventory levels.
Defect Rate	Average defect rate is **9.98%**, showing scope for quality improvement.	Improve quality control and reduce defects.
Product Sales	**Cosmetics** has the highest sales at **2.38M units**.	Focus inventory planning on high-demand products.
Lead Time	Average supplier lead time is **17.45** time units.	Monitor delays and improve supply planning.
Data Scope	Dataset contains **15,000 records** and is project-based.	Use larger real-world datasets for future analysis.
Dashboard	Dashboard provides high-level supply chain insights.	Add detailed pages for deeper analysis.

8. Conclusion and Recommendations

8.1 Conclusion

The SupplyChain360 Power BI project successfully converts supply chain data into an interactive dashboard.

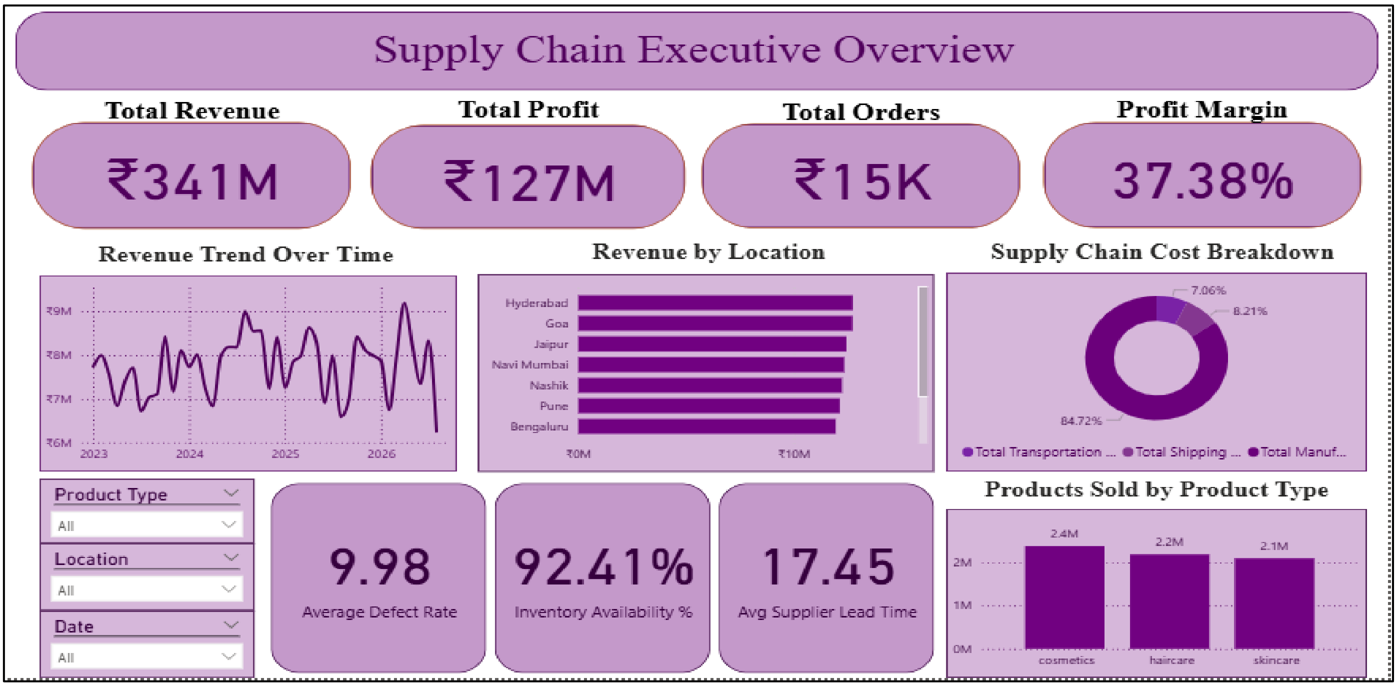
The project uses a Star Schema with one fact table and three dimension tables: Product, Location, and Date. DAX measures were used to calculate important KPIs.

The dashboard shows ₹341M revenue, ₹127M profit, 15K transactions, and 37.38% profit margin. Overall, the project demonstrates how Power BI can support supply chain analysis and data-driven decision-making.

8.2 Recommendations

- **Reduce Manufacturing Costs:** Focus on the major **84.72%** cost contribution.
- **Improve Product Quality:** Reduce the **9.98%** defect rate.
- **Monitor Lead Time:** Track supplier lead time to avoid delays.
- **Maintain Inventory:** Maintain the **92.41%** availability level.
- **Focus on High-Demand Products:** Give more attention to **Cosmetics**.
- **Analyze Locations:** Monitor location-wise revenue performance.

DashBoard:



Project Summary

What did we create?

We created a Supply Chain Analytics project in Power BI using the SupplyChain360 integrated supply chain dataset. The project analyzes sales, inventory, manufacturing, product quality, locations, and overall supply chain performance through interactive dashboards and KPIs.

Why did we create it?

The purpose of the project is to convert raw supply chain data into meaningful business insights. The dashboard helps identify revenue trends, cost patterns, inventory availability, product demand, lead times, quality issues, and operational performance.

How did we create it?

The original supply chain dataset was expanded into 15,000 transaction records and organized into a relational Star Schema containing one fact table and three dimension tables: Product, Location, and Date. The data was imported into Power BI, relationships were created, DAX measures and KPIs were developed, and interactive visualizations and slicers were added to create the final Executive Overview dashboard.

The completed project demonstrates the practical use of data modelling, DAX, KPIs, Business Intelligence, and interactive visualization for supply chain decision-making.